MAT1375, Classwork10, Fall2026

Ch10. Rational Functions I

1. Definition of the **Rational function**:

A rational function is a fraction of two polynomials $f(x)=\frac{p(x)}{g(x)}$, where $p(x)$ and $g(x)$ are both polynomials, and $g(x)\neq 0$.

The **domain of a rational function** f is all real numbers for which the denominator $g(x)$ is not zero:

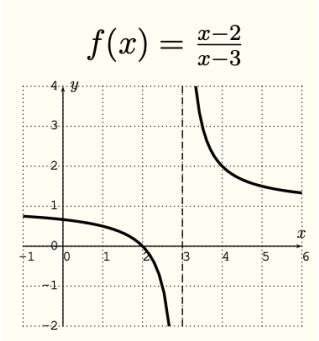
$$D_f=\{\,x\in\mathbb{R}\mid x\in D_p\cap D_g \text{, but } g(x)\neq 0\,\}$$

2. **Arrow Notation**: Given a constant a and we have

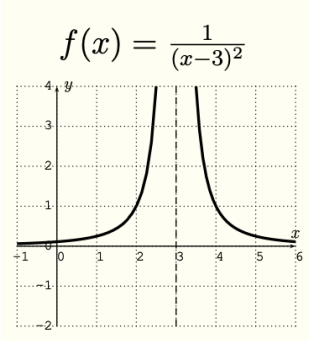
$x\to a^+$:	x approaches a from the **right** (x is very closed to a but $x\neq a$ and $x > a$)
$x\to a^-$:	x approaches a from the **left** (x is very closed to a but $x\neq a$ and $x < a$)
$x\to\infty$:	x approaches infinity (x increases without bound)
$x\to-\infty$:	x approaches negative infinity (x decreases without bound)

3. The definition of a **Vertical Asymptote**:

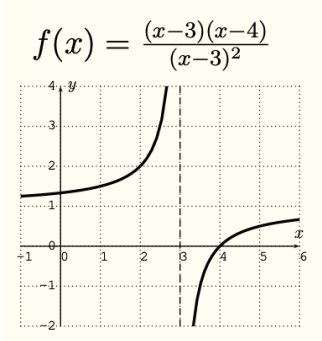
The line $\boldsymbol{x=a}$ is a _____ of the graph of a function f if $f(x)$ **increases or decreases without bound** as $\boldsymbol{x}$ **approaches** $\boldsymbol{a}$. Here are three examples:



As $x\to 3^+$, $f(x)\to\infty$;
As $x\to 3^-$, $f(x)\to-\infty$.



As $x\to 3^+$, $f(x)\to\infty$
As $x\to 3^-$, $f(x)\to\infty$



As $x\to 3^+$, $f(x)\to-\infty$
As $x\to 3^-$, $f(x)\to\infty$

4. How to locate Vertical Asymptotes: Let $f(x)=\frac{p(x)}{g(x)}$ be a rational function.

(1) If $p(x)$ and $g(x)$ have no _____, and $\boldsymbol{a}$ is a **zero** of $g(x)$ which makes $f(x)$ _____, then _____ is a vertical asymptote of the graph of $f(x)$.

(2) If $\boldsymbol{a}$ is a **zero** of both $p(x)$ and $g(x)$ ($p(a)=$_____, $g(a)=$_____.) which means _____ is the common factor of $p(x)$ and $g(x)$, then there is a _____ at $x=a$ and there is _____ at $x=a$.

5. Find the vertical asymptotes of the graph of each rational function:

a) $f(x) = \frac{x}{x^2-1}$ b) $g(x) = \frac{x-1}{x^2-1}$ c) $h(x) = \frac{x-1}{x^2+1}$